



UAEH

Enfermedad Renal



2026

User manual

Manual version

Version	Date	Author	Changes
1.0	01/07/2026	Ing. Jocsan Uziel Martínez López	Document creation

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Introduction

This manual aims to guide the user in using the software for preprocessing data on Chronic Kidney Disease (CKD). Throughout the document, the requirements and procedures for correctly running the application and obtaining the expected results are described.

The preprocessing software was developed to run using Jupyter Notebook and is composed of several sequential processes that allow the preparation of patient datasets. During proper execution, the following functions are performed: information cleaning, treatment of inconsistent or incomplete data, and reduction of biases that influence the results obtained.

This helps to improve the quality and reliability of the data that will be used in later stages.

To complete the process correctly, it is necessary to run the Jupyter files in the specified order. Each file contributes a specific function within the workflow, so each one fulfills an essential step for the process to run smoothly.

Upon completion, the system will generate a CSV file containing the data set to be processed in the next stage.

System requirements

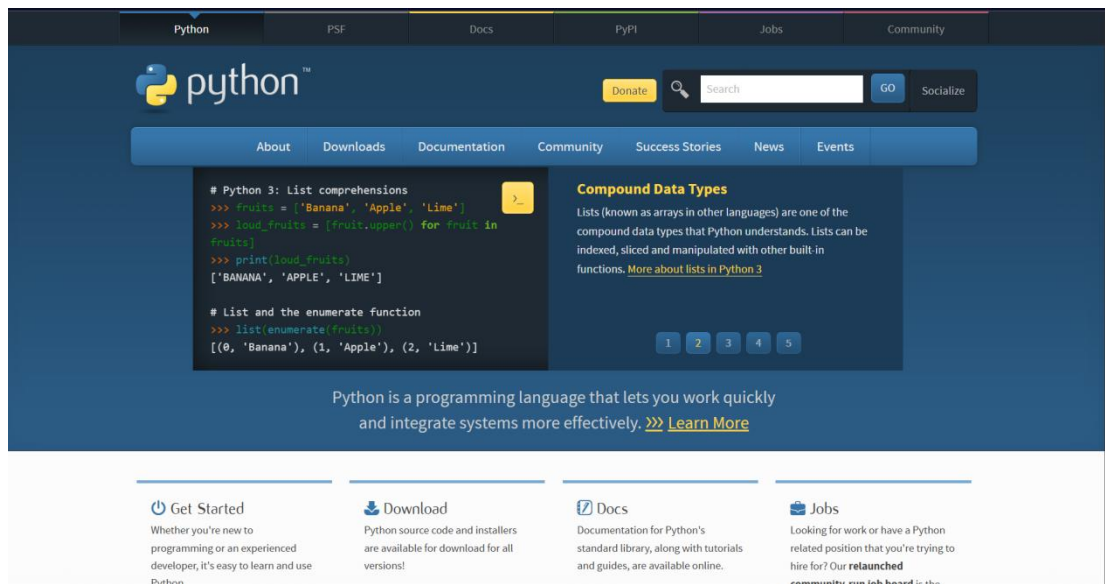
Python The

Python programming language It is an essential requirement for the software to function, as it is responsible for interpreting and executing the files related to data preprocessing. Therefore, it is necessary to have the language installed on the system and integrated into the environment variables so that it can be correctly recognized by the computer.

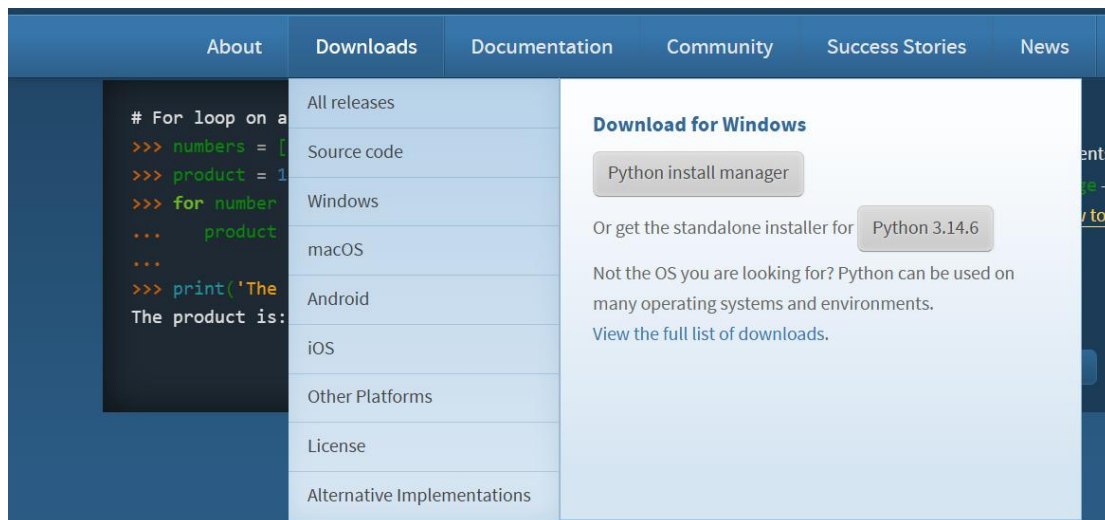
The following images describe the steps required to install and configure Python appropriately.

<https://www.python.org/>

First, you must open the link above. Once opened, you will see a screen like the image below, which is the main screen of the official Python website.



Download option is located on the home screen , allowing access to the available installation versions for different operating systems. This manual describes the download and installation procedure for **Windows** ; however, the process is similar for other compatible operating systems.



Once you select the **Download option** and then **Windows** , the browser will redirect you to the page where the different versions of the Python programming language are available for download.

As of the date of this manual's creation, the latest stable version available is **3.13.14** . It is recommended to download the file corresponding to the latest available version that contains the name " **Installer** ," as this offers greater stability, compatibility, and support for running the software.

Python Releases for Windows

- [Latest Python install manager - Python install manager 26.3](#)
- [Latest Python 3 Release - Python 3.14.6](#)

Stable Releases

- [Python install manager 26.3 - June 30, 2026](#)
 - [Download Installer \(MSIX\)](#)
 - [Download MSI package](#)

- [Python 3.13.14 - June 10, 2026](#)

Note that Python 3.13.14 cannot be used on Windows 7 or earlier.

- [Download Windows installer \(64-bit\)](#)
- [Download Windows installer \(32-bit\)](#)
- [Download Windows installer \(ARM64\)](#)
- [Download Windows embeddable package \(64-bit\)](#)
- [Download Windows embeddable package \(32-bit\)](#)
- [Download Windows embeddable package \(ARM64\)](#)
- [Download Windows release manifest](#)
- [Python 3.14.6 - June 10, 2026](#)
 - [Download using the Python install manager.](#)

Depending on your computer's processor architecture, you will need to select either the **64-bit** or **32-bit installer** . Most modern computers use a **64-bit architecture** , so in most cases, this will be the correct option.

When you select the installer, your browser will display a window asking for the location where you want to save the executable file. Choose your preferred destination folder and confirm the download to begin the process.

Note that Python 3.13.14 cannot be used on Windows 7 or earlier



- Download [Windows installer \(64-bit\)](#)
- Download [Windows installer \(32-bit\)](#)
- Download [Windows installer \(ARM64\)](#)
- Download [Windows embeddable package \(64-bit\)](#)
- Download [Windows embeddable package \(32-bit\)](#)
- Download [Windows embeddable package \(ARM64\)](#)
- Download [Windows release manifest](#)
- [Python 3.14.6 - June 10, 2026](#)
 - [Download using the Python install manager.](#)

Once the download is complete, the installer executable file will be located in the previously selected location. Double-click on it to start the installation wizard.

During the process, follow the wizard's instructions and continue with the installation using the default options. Before starting the installation, ensure that the **"Add Python to PATH" checkbox** is selected. This option automatically adds Python to your system's environment variables, making it easier to run from the command line and ensuring the software functions correctly.



Once the installation process has completed successfully, click **Finish** to close the installation wizard. Python will then be installed and configured on your system, allowing you to continue with the following steps described in this manual.

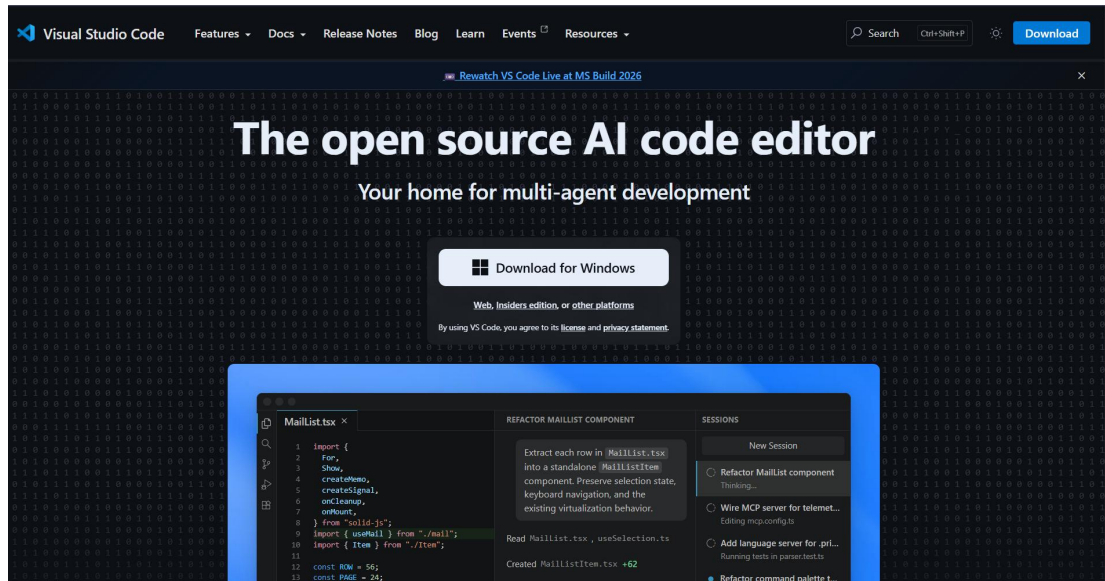
VS Code

<https://code.visualstudio.com/>

To continue with the installation of the development environment, you need to download **Visual Studio Code** . The provided link will redirect you to the main download page, as shown in the image below.

By default, the site detects the computer's operating system and displays the corresponding version as the primary option. This manual uses the **Windows version** ; however, installers for other operating systems are also available.

By selecting the **Download for option On Windows** , the browser will download the installer executable file to the location previously selected by the user.

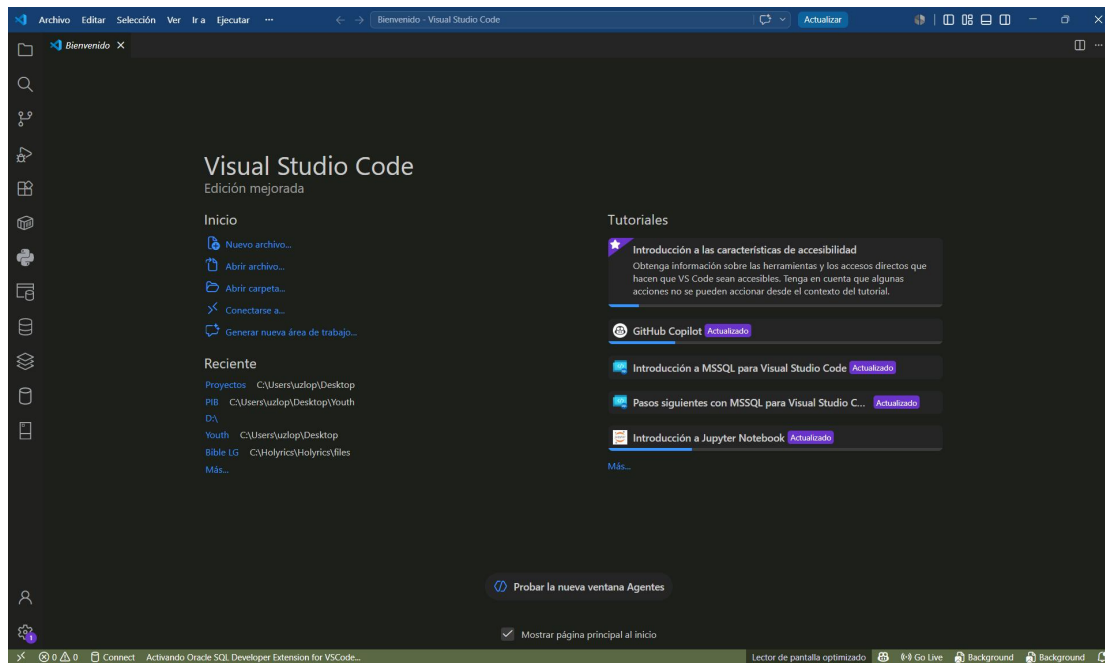


Once the installation file has downloaded successfully, double-click it to run the installer. As with the **Python installation**, the **Visual Studio Code** installation wizard will open .

Follow the on-screen instructions and complete each step of the wizard until the installation is finished. It is recommended to keep the default options unless there are specific configuration requirements.



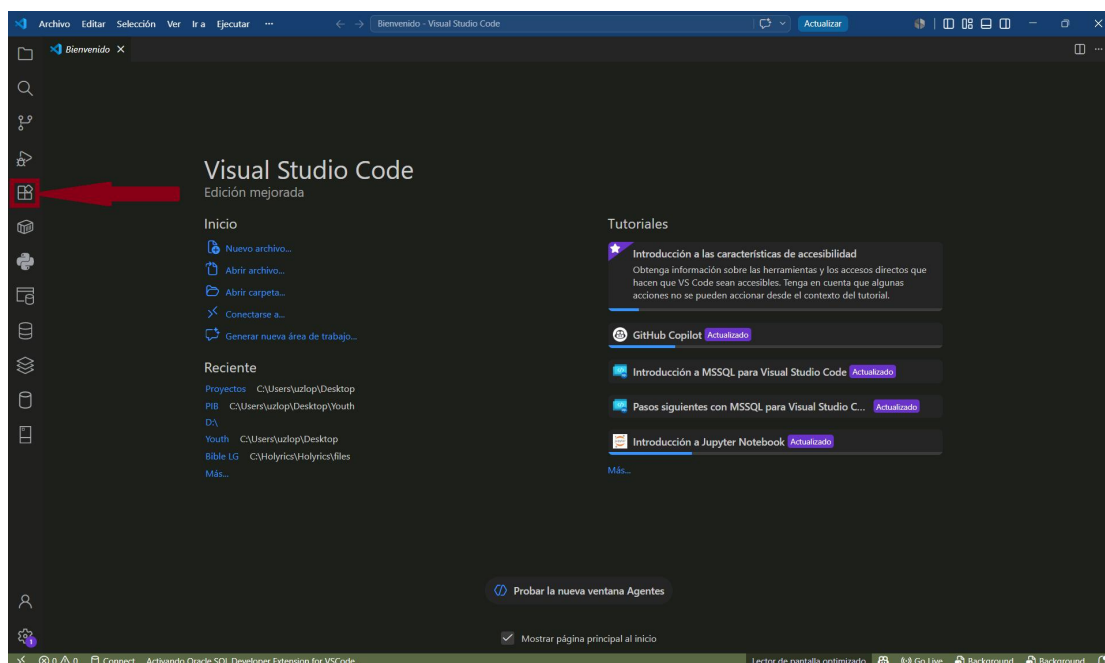
Once the installation is complete, **Visual Studio Code** will start automatically and display a window like the one shown in the following image. From this environment, you will be able to open, view, and run the code files that make up the system described in this manual.



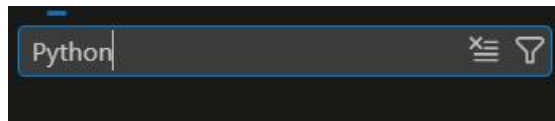
Python extension

Once **Visual Studio Code** is installed , you will need to select the **Extensions** icon located in the left sidebar, as shown in the following image. From this section, you will be able to install the **Python extension** .

Installing this extension is essential, as it allows you to integrate the **Python language** with **Visual Studio Code** , providing the necessary tools to run, debug, and manage project files properly.



Once inside the **Extensions section** , a search bar will appear. Type "**Python**" in it to locate the corresponding extension. Then, select the official Python extension and click "**Install**" to begin the installation.



When you perform the search, extensions related to the term **Python** will be **displayed** . It is recommended to install the **Python** and **Python Debugger** extensions , as they provide the necessary tools for running, debugging, and developing projects written in this language within **Visual Studio Code** .

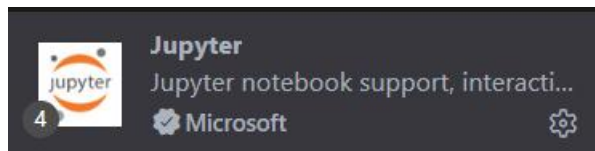


Jupyter extension

Similar to installing the **Python extension** , you'll need to use the search bar in the **Extensions section** to locate the **Jupyter extension** . Once found, select and install it to enable the creation, opening, and running of **Jupyter Notebook files** within **Visual Studio Code** .



Once you locate the **Jupyter extension** , click on it and select the **Install option** to begin the installation. This extension is essential, as it allows you to open and run **Jupyter Notebook (.ipynb) files** within **Visual Studio Code** , displaying the code, results, and visualizations in an interactive notebook format, which facilitates development and data analysis.



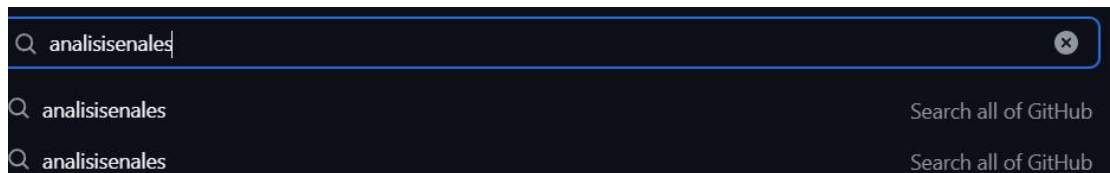
Download and save

Download rar

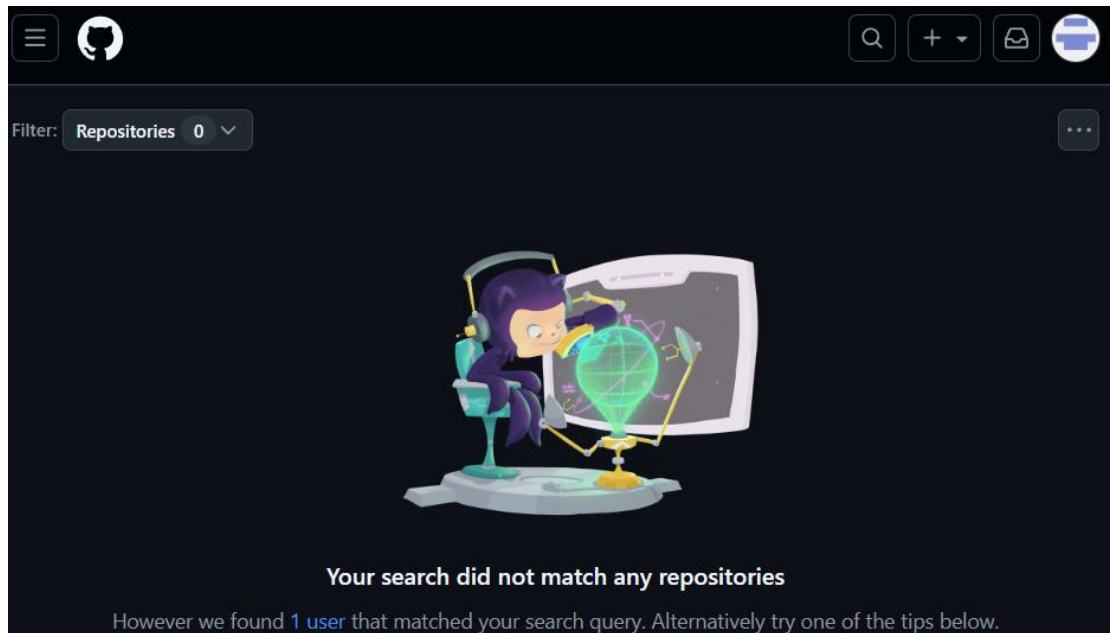
To download the project in **.zip** or **.rar format** from **GitHub**, you need to log in with a user account on the platform. Once logged in, you can access the repository and download the files necessary to run the system.



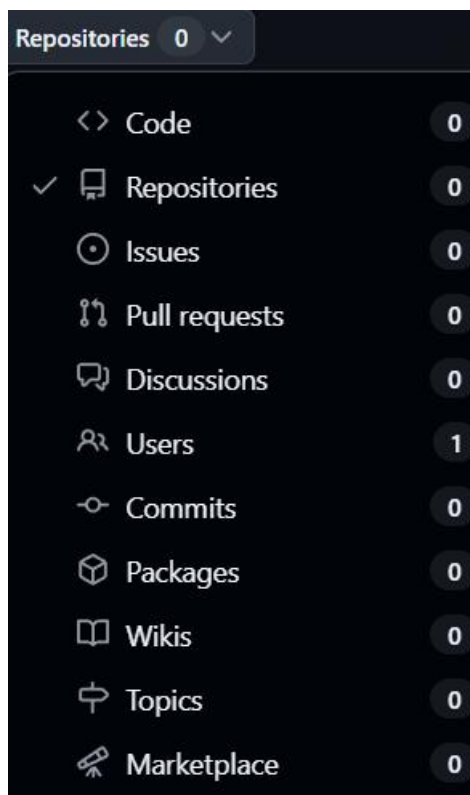
Once you're logged into **GitHub**, use the search bar at the top of the page to find the repository. Type "**anlisisenales**" and select the corresponding repository from the search results to access its content.

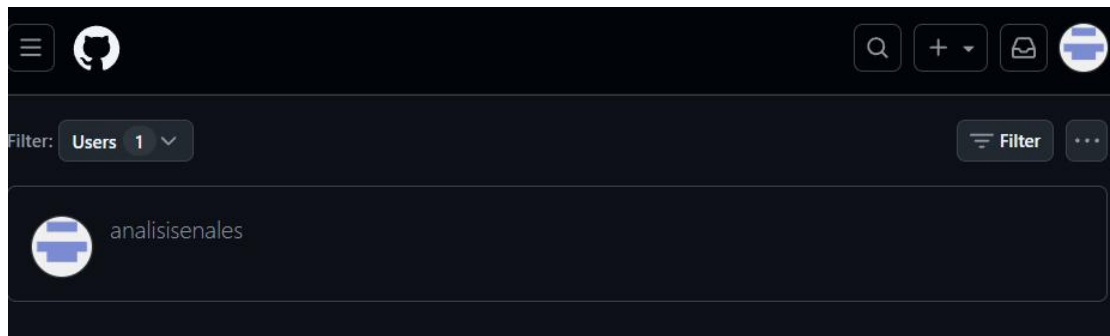


A screen similar to the one shown in the following image will then appear. The appearance of the results may vary depending on the type of search performed or the filters applied in **GitHub**; however, this is normal behavior and does not pose any problem for continuing with the procedure.

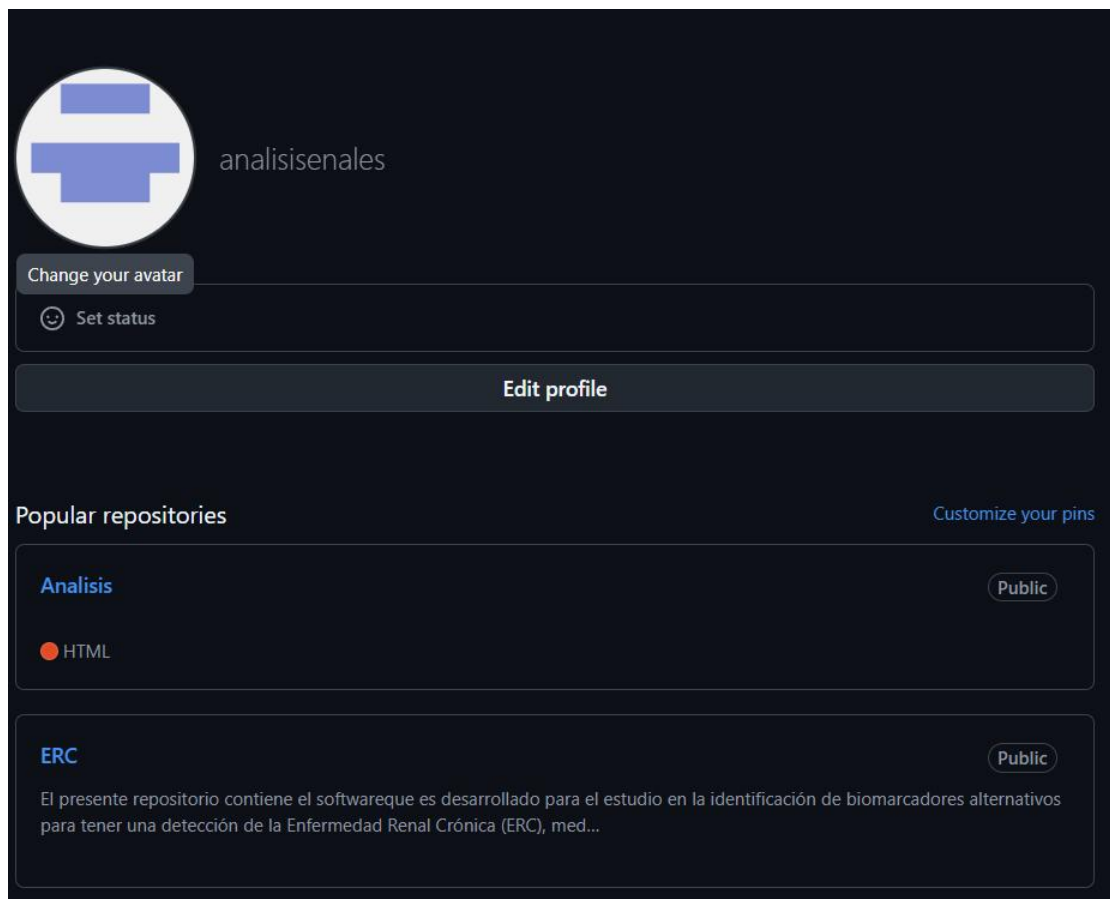


In the **Repositories** section , click on the search filter to expand the available categories. Then, select the **Users** option to limit the results to only registered users on the platform who match your search .

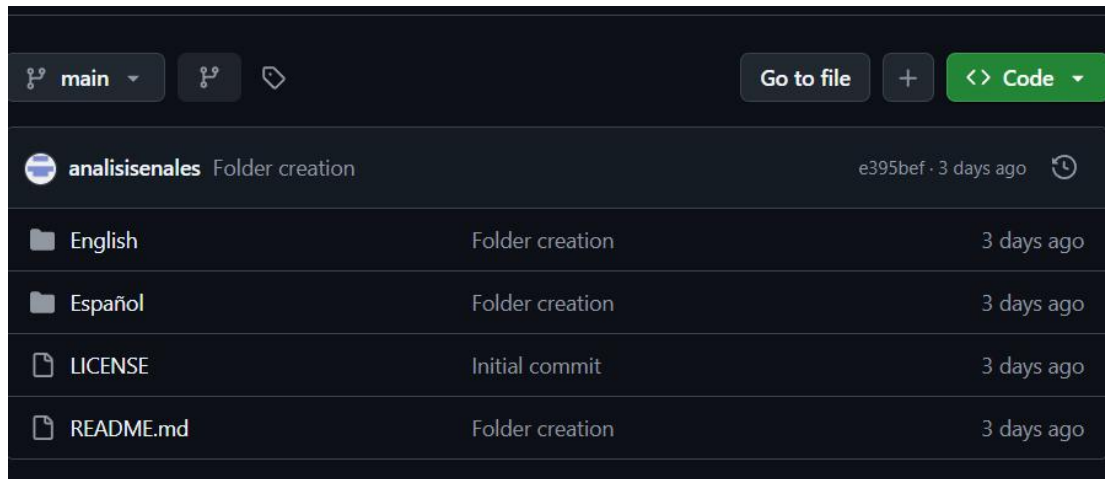




Selecting the corresponding option will display the user matching the search criteria. Accessing their profile will open a new section showing the available repositories associated with that account.

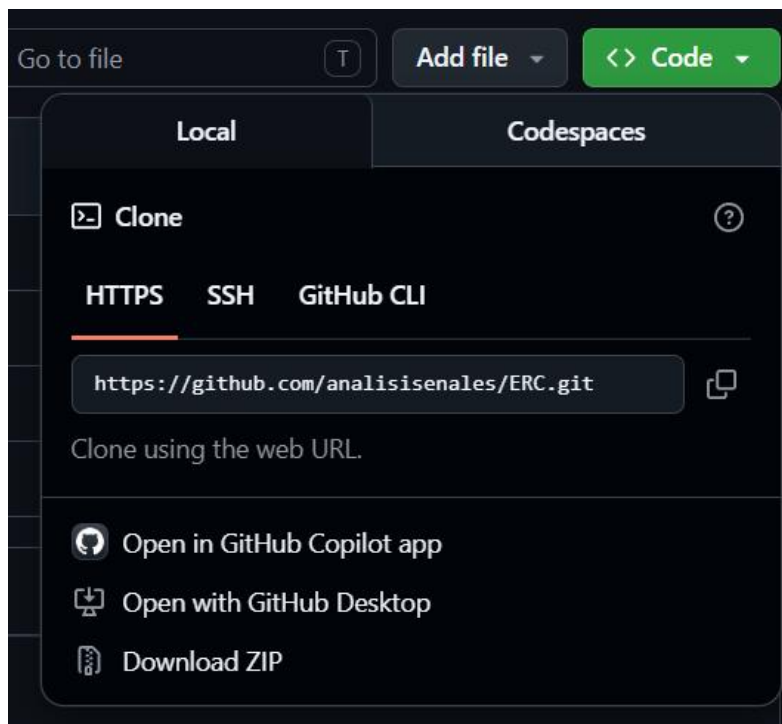


Select the repository named **ERC** to access its contents. This section will display the files and resources that make up the project, which will be used to continue with the system configuration and execution process.



If you need to download the entire repository, go to the green "**Code**" button located within the repository. Clicking this button will display a menu with different download options, as shown in the following image.

Select the **Download ZIP** option to obtain a compressed copy of the repository and be able to use the project files on your local computer.

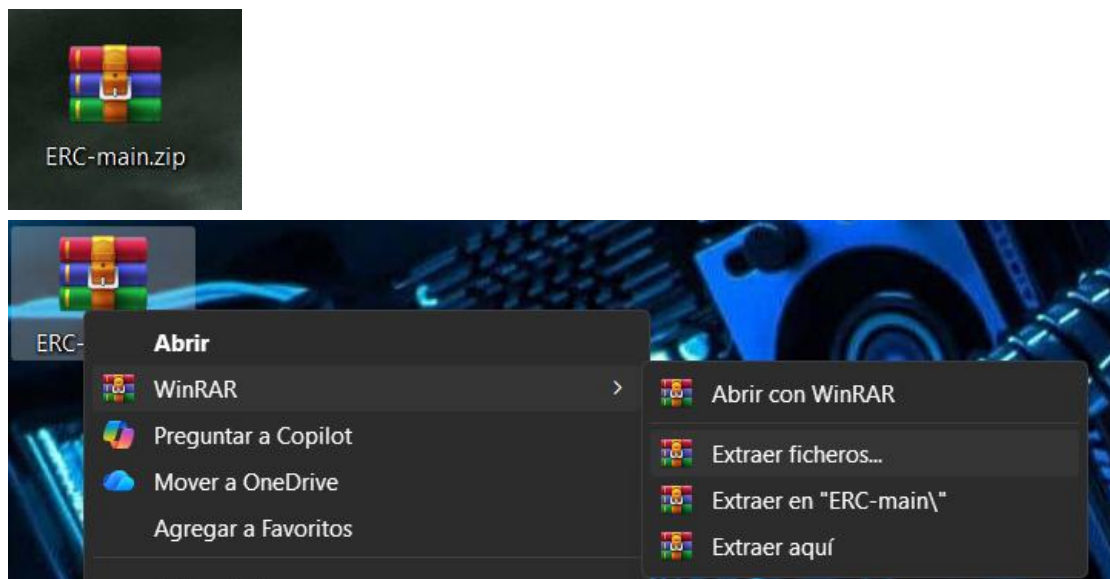


After selecting the download option, your browser will prompt you to choose the location where you want to save the compressed repository file. Select your preferred destination folder and confirm the download to store the project files on your local computer.

Unzip rar

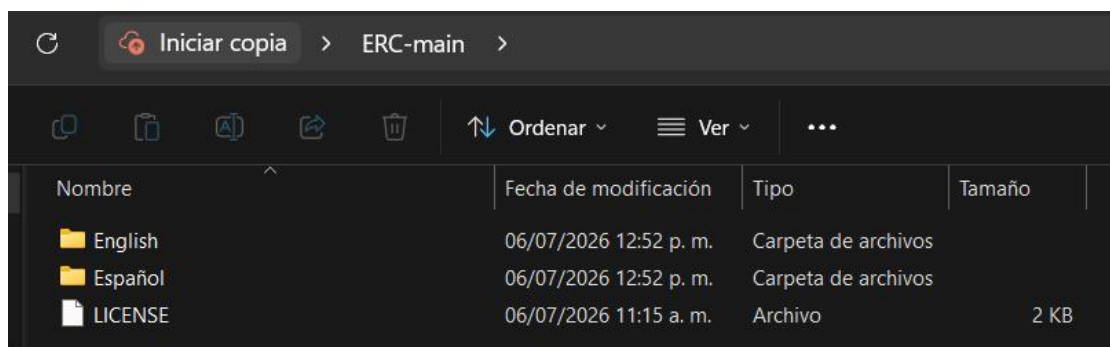
Once the repository has been downloaded from the **GitHub platform** , the resulting file will contain the elements necessary for the software to function.

To access the project files, you will need to extract the compressed file. Right-click on the downloaded file, and **WinRAR** will display the extraction options .



"**Extract here**" option , the file will be decompressed and a new folder will be created with the files corresponding to the project.

Within this folder, you will find two subfolders organized by language: "**Spanish**" and "**English**" . Each subfolder contains the user manual in its respective language, allowing you to consult the software documentation as needed.



Inicio > ERC-main > English >

Nombre	Fecha de modificación	Tipo	Tamaño
assets	06/07/2026 11:15 a. m.	Carpeta de archivos	
Preprocessing	06/07/2026 11:15 a. m.	Carpeta de archivos	
Processing	06/07/2026 11:15 a. m.	Carpeta de archivos	
README.md	06/07/2026 11:15 a. m.	Archivo de origen ...	1 KB
User manual.docx	03/07/2026 11:30 p. m.	Documento DOCX	4,546 KB

Inicio > ERC-main > Español >

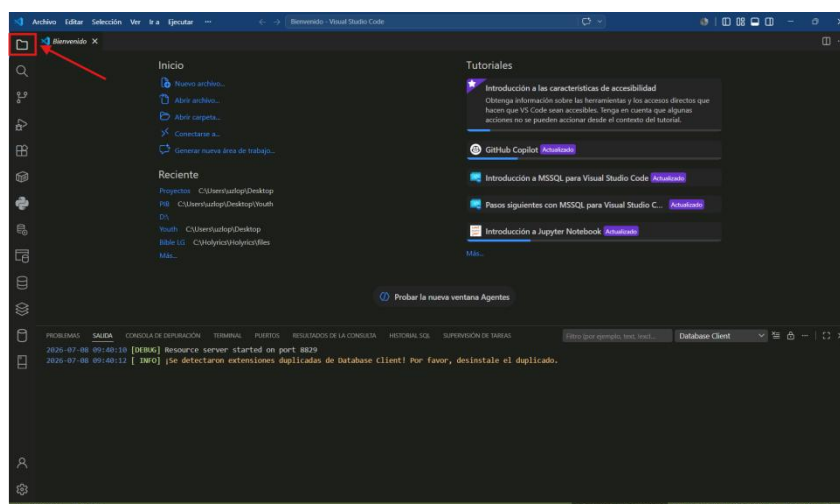
Nombre	Fecha de modificación	Tipo	Tamaño
assets	06/07/2026 11:15 a. m.	Carpeta de archivos	
Preprocesamiento	06/07/2026 11:15 a. m.	Carpeta de archivos	
Procesamiento	06/07/2026 11:15 a. m.	Carpeta de archivos	
Manual de usuario.docx	03/07/2026 11:30 p. m.	Documento DOCX	4,546 KB
README.md	06/07/2026 11:15 a. m.	Archivo de origen ...	1 KB

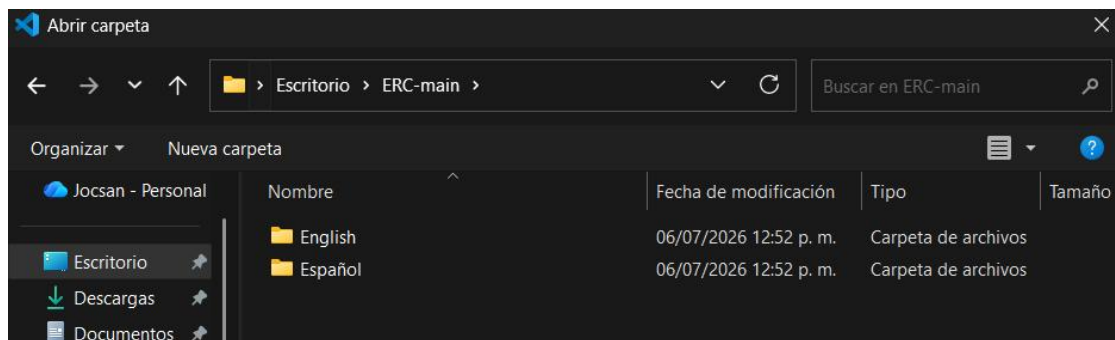
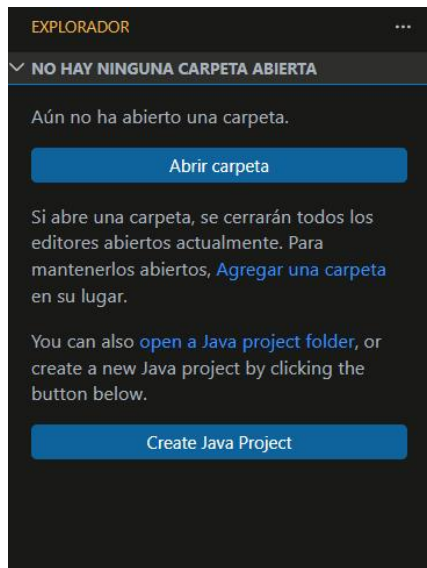
Interface

Modules

To run the project files, you will need to open the **Visual Studio Code application** . Once inside, go to the left sidebar and select the **File Explorer icon** , which looks like a folder.

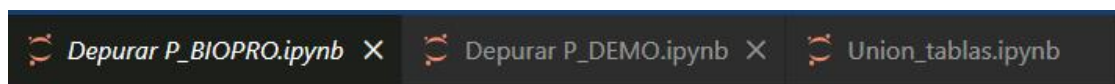
Clicking this button will open a menu with various options, including the option to **open a folder** . This function will allow you to load the previously extracted project folder to continue configuring and running the corresponding files.





By selecting the folder corresponding to your preferred language, you will be able to access the project structure within **Visual Studio Code** .

To begin the process, you must enter the **Preprocessing folder** , where the files responsible for debugging and processing the datasets used by the software are located.



Data files

To correctly use the files corresponding to the **preprocessing process** , it is necessary that the data sets are in **.xpt format** .

The software is designed to work with this type of file, so if the data is in a different format, the execution and debugging processes will not be able to be performed correctly.



Modular implementation

The execution of the **preprocessing modules** must be carried out following the established sequence, since each file generates information necessary for the operation of the next module.

The execution order is as follows:

- Debug P_BIOPRO.ipynb
- Debug P_DEMO.ipynb
- Union_tablas.ipynb

Features

Creating CSV files

Once the previously described execution sequence is complete, the resulting files will be generated in **CSV format**. These files contain the information processed during the preprocessing stage and will be essential for the next phase of the project.

The generated files will be used as input for subsequent data analysis and processing using the complementary tools that make up the software workflow.

